Conference Proceedings

Design, Startup, and Operation of a 50 MWe CO_2 Sequestration System

Abstract 351

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1.0 Abstract

The Midwest Geological Sequestration Consortium, one of seven U.S. Department of Energy Regional Sequestration Partnerships, began injection of 1,000 tonnes (metric tons) per day of CO_2 at the Archer Daniels Midland facility in Decatur, Illinois in November 2011. This is known as the Illinois Basin – Decatur Project (IBDP). Trimeric Corporation had process engineering design responsibility for the compression and dehydration facility that is used to take atmospheric pressure CO_2 from the ADM ethanol fermentation process and deliver dehydrated CO_2 to the injection well. An overview of the project design basis and the requirements that led to the selected process design will be presented. An overall system design review including the multistage centrifugal blower, reciprocating compressors, triethylene-glycol dehydration system, multistage centrifugal pump and the pipeline will be presented. Lessons learned during the startup and commissioning phases will be presented. A comparison of this facility to compression and dehydration facilities for a commercial scale 500 MWe facility will be presented.

2.0 Project Background

The Illinois Basin - Decatur Project is a large scale CO_2 sequestration demonstration project, led by the Midwest Geological Sequestration Consortium and funded by the U.S. Department of Energy. The host site is the Archer Daniels Midland facility in Decatur, Illinois. The scope of the project is to compress and dehydrate 1,000 tonnes / day of 99% + pure CO_2 from ethanol plant fermenters and inject the CO_2 in a saline aquifer 7,000 feet underground. Injection began in November 2011 and will continue for nominally three years until one million tonnes total have been injected. The MGSC will monitor CO_2 subsurface plume migration, groundwater, and surface conditions during the three-year injection period and for three years after injection stops.

3.0 Process Design Basis and Permit Conditions

This section discusses the process design basis and injection permit requirements for the project.

The process design basis included the functional design requirements of the surface facilities for the compression and dehydration of the carbon dioxide for injection.

The CO_2 flow delivery requirement was 1 million tonnes (dry basis) of CO_2 over a three-year period. The surface facilities were designed for 24-hour operation, with no more than 30 days of downtime per year. Downtime includes scheduled downtime and unscheduled downtime. These requirements led to a minimum specified design rate of 995 tonnes per day for the compression and dehydration equipment.

The project timeline required ordering compression and dehydration equipment without knowing the surface injection pressure that would be required to meet the desired injection rate. At the time the equipment was ordered, the injection well had not been permitted or drilled and predictions of required surface injection pressures made by project geologists and reservoir engineers were made based on data from the nearest known well, which was several miles away. At the time of project initiation, the state of knowledge in defining the surface pressure

required for injection was based on model data shown in Figure 1. The graph shows the relationship between down-hole pressure and fluid injection rate in barrels per day with an over-lay of constant surface pressure and injectivity curves. The injectivity index (PI) quantifies the pressure increase due to pumping a known rate and volume of fluids into a formation. At a given injectivity index, the down-hole pressure increases as the injection flow increases. At a higher formation injectivity value, the increase in down-hole pressure is less steep with an increase in fluid flow, thus more flow to the formation. Note that 1,100 tonnes per day is approximately 10,000 bpd.

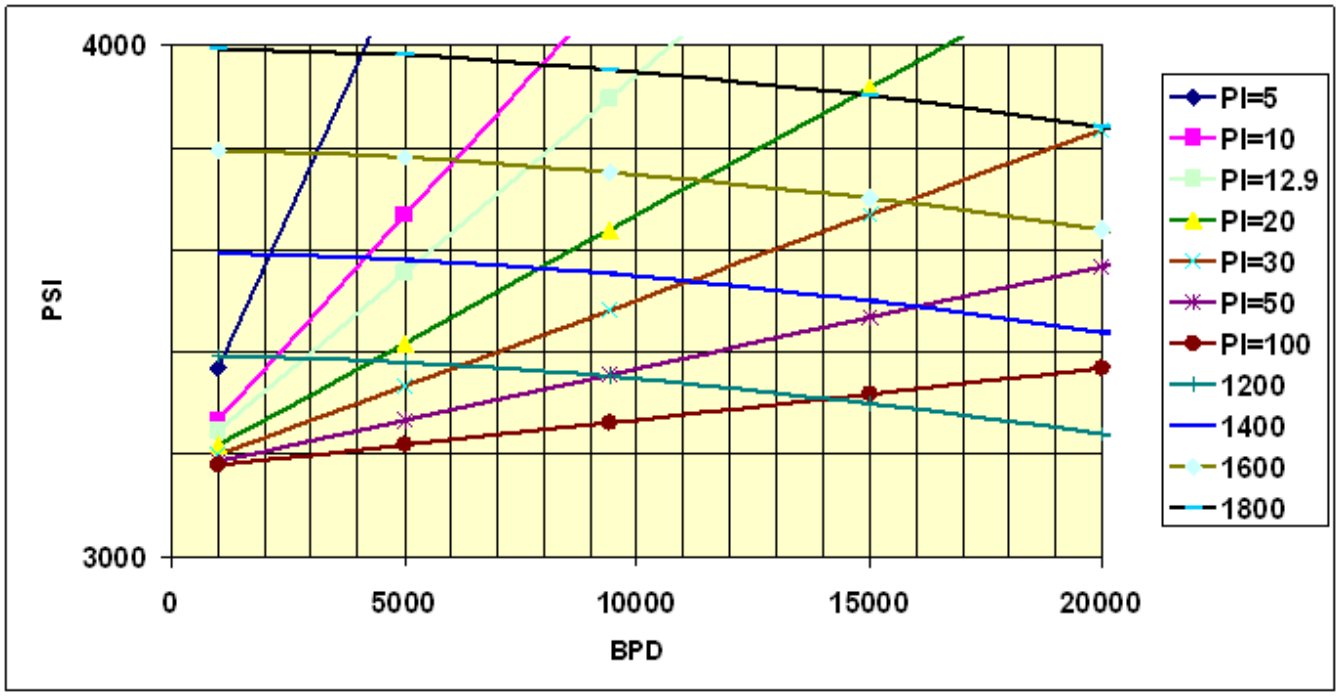


Figure 1 – Flow vs. Down-hole Pressure as a Function of Injectivity (PI) and Surface Pressure

During initial injection into the Mt. Simon formation, it was anticipated that due to displacement of water in the formation by the CO_2, the injectivity will resemble that of 5 to 20 PI curves and that after a large amount of water is displaced, the injectivity will follow the 20 to 100 PI curves. Based on this data, it was believed that a CO_2 injection rate of 1,100 tonnes per day into the Mt. Simon formation was achievable with selection of compression equipment capable of operating at surface injection pressures ranging from 1,400 to 1,950 psig.

The original CO_2 delivery requirements for injection rate, pressure, temperature, and phase condition are summarized in the Table 1.

Table 1: CO_2 Delivery Requirements

Operational Parameter	Project Design Requirement
Injection Rate	Able to inject at rates ranging from 250 – 995 tonnes/day
Flow Control	Maintain within 10 % of target flow
Maximum Wellhead Inlet Pressure	1,350 – 2,000 psig
Minimum Wellhead Inlet Pressure	During startup: 0 psig – 1,057 psig, During normal operations: 1,057 psig.
Pressure Control	Injection pressure is not controlled when the injection rate is specified
Maximum Wellhead Inlet Temperature	120 ^oF
Minimum Wellhead Inlet Temperature	88 ^oF
Temperature Control	Target set point after reciprocating compressors was 95 ^oF. Not controlled if multistage centrifugal pump is used, which adds 10 ^oF to 15 ^oF to CO_2 surface temperatures.

Each of the specifications in Table 1and their rationale are discussed below.

INJECTION RATE

The system was required to deliver CO_2 to the injection well at any specified flow rate from 250 – 995 tonnes per day. This turndown capability was specified in order to allow the flexibility to conduct injection experiments over a wide range injection rates. The system was designed for a maximum capacity of 1,100 tonnes per day to add a 10% safety factor to the equipment design.

FLOW CONTROL

The system was required to maintain CO_2 flow to within +/- 10 % of the targeted flow rate. For example, for steady-flow test at an injection rate of 500 tonnes per day, the system must be able to control the flow to between 450 and 550 tonnes per day during that test. This range was set based on expected typical performance of compression systems operating at steady flow conditions and typical accuracy and performance of standard flow control equipment.

MAXIMUM DELIVERY PRESSURE

The system was to be capable of delivering CO_2 at a maximum surface pressure between 1,350 and 2,000 psig at the wellhead. At the time of initial design and equipment selection process, the actual required injection pressure was unknown.

MINIMUM CO_2 DELIVERY PRESSURE

During startup, the system was to be designed to accommodate the increase in injection wellhead inlet pressure from 0 psig during initial startup up to the expected maximum operating surface pressure during normal operation. As specified in the draft Underground Injection Control, UIC injection permit, it was required that during normal operation, the system had the capability of delivering CO_2 to the wellhead at a minimum pressure of 1,057 psig (or 1,071 psia, which is the critical pressure of CO_2).

WELLHEAD PRESSURE CONTROL

Since pressure of the CO_2 delivered to wellhead is generally determined by characteristics of the reservoir and the injection rate, there was no design requirement to maintain control over surface injection pressure beyond meeting the specification for minimum and maximum pressure as noted above. At steady-state operations and barring any changes in injection rate, the wellhead pressure should remain relatively constant. The system operating procedures were developed to minimize rapid pressure changes during startup and shutdown to less than 50 psig per minute. It may not be possible in all cases to meet this requirement though. For example, if there is a sudden loss of power at the plant and the compressors shut down, the wellhead pressure could drop rapidly, and the pressure drop could exceed 50 psig per minute.

MAXIMUM CO_2 DELIVERY TEMPERATURE AT INJECTION WELLHEAD

A maximum wellhead temperature of 120 ^oF was specified. This is a typical summer-condition, outlet temperature for air-cooled heat exchangers, commonly used in gas compression. Using water-cooled exchangers will likely result in lower surface injection temperatures year round. The final Underground Injection Control (UIC) permit had an upper limit of 150 ^oF for the CO_2 delivered to the inlet to the injection well.

MINIMUM CO_2 DELIVERY TEMPERATURE AT INJECTION

A minimum wellhead temperature of 88 ^oF in the design basis was based on parameter limits specified in the initial draft of the UIC permit. It was noted that during a cold-weather start-up and during normal operations prior to insulation of the above ground CO_2 pipeline that the CO_2 could be colder than 88 ^oF at the wellhead. Subsequent revisions to the UIC permit were based on a minimum surface injection temperature requirement of 60 ^oF. The final UIC permit required injection of CO_2 in the supercritical state at the point of injection into the formation, which means that the CO_2 must be above its critical temperature of 88 ^oF at the point of injection into the formation. Modeling over the range of expected surface pressures, showed that with heating due to hydrostatic compression of CO_2 in the injection well, 60 ^oF CO_2 delivered at the inlet to the wellhead would result in down-hole temperatures greater than 88 ^oF. This change in the UIC permit allowed compliance to be monitored with surface pressure and temperature gauges, which are more accessible for calibration and service than down-hole instruments.

TEMPERATURE CONTROL

Other than the minimum and maximum temperature limits listed above, there was no requirement to maintain or control the CO_2 temperature at the wellhead at a specific set point. It was anticipated that the CO_2 temperature at the wellhead inlet would remain relatively constant with consistent operation of cooling equipment associated with the compression system.

In addition to these operational parameters, Table 2 includes the purity specifications for the injection CO_2 stream.

Table 2: Injection CO_2 Purity Specification

Component	Purity Specification
Carbon Dioxide	Minimum 99 %
Oxygen	100 ppmv maximum
Water	30 lb/MMscf (633 ppmv) maximum

Each of the specifications in Table 2 and their rationale are discussed below.

CARBON DIOXIDE

The fermenter vent gas from an ethanol plant is generally high purity CO_2 (> 99% vol.). The source point specified for this project was downstream of an existing water scrubbing system, thus there would be minimal (ppm level) impurities from ethanol production remaining in the CO_2 to be compressed and dehydrated prior to injection.

OXYGEN

Typical analysis of the source stream indicated that normal oxygen content was below 10 ppmv. A measurement of oxygen significantly higher (10x) than this level would indicate ingress of air from a possible loss of pressure control on a fermenter.

WATER

As stated above, the source point for the CO_2 is down stream of an existing water scrubber and therefore would be considered water saturated. A typical CO_2 pipeline specification of 30 lb/MMscf was selected in order to prevent the formation of free water or hydrates in the transfer pipeline to the wellhead.

4.0 Equipment Screening and Selection Process

4.1 Equipment Screening

A study was conducted to identify the preferred equipment configuration for this project to compress the ethanol fermenter vent stream of CO_2 from approximately 1 psig to a maximum of 2,000 psig at the wellhead.

COMPRESSION CONFIGURATIONS

The following is a list of the equipment combinations considered for CO_2 delivery:

- Blower, Screw Compressors, Reciprocating Compressors
- Blower, Reciprocating Compressors
- Blower, Reciprocating Compressors, Pump
- Blower, Screw Compressors, Reciprocating Compressors, Pump
- Compression, Liquefaction, Pump

The configurations were compared using budgetary purchased equipment costs (+/- 20% accuracy) and the estimated power consumption over a 3-year project life. In addition, each of the configurations utilizing a blower was evaluated without a blower in order to ascertain the impact of the blower on project costs.

The estimated purchased equipment costs for configurations utilizing screw compressors downstream of a blower and upstream of reciprocating compressors were on average 32% less than blower/reciprocating compressor configurations. However, the 3-year electrical costs of the configurations that included a screw compressor were on average 8% more than blower/reciprocating compressor configurations. The total estimated purchased equipment plus 3-year electrical costs for compression equipment configurations that did not utilize screw

compressors (reciprocating compression only) were on average 6% higher than the options that included screw compressors.

Excluding the blower from an equipment configuration significantly increased estimated purchase cost of the subsequent compression equipment by an average of 35% and increased the 3-year electrical cost by an average of 4%.

Budgetary quotes were obtained for a compression to 800 psig, followed by condensation using chilled water and then pumping to the final injection pressure. This approach showed no clear purchased equipment or operating cost advantage and was judged to have more complexity than compression based options.

As a follow up to the options studied above, Trimeric compared a refrigeration based CO_2 liquefaction and pumping approach to the compression approach. The two processes were simulated assuming inlet conditions of 80 ^oF and 250 psig (the compression from 0 to 250 psig would be the essentially the same on this project for both approaches). The results showed that the liquefaction option had a power consumption that was 38% greater than the reciprocating compressor option. The three year injection period of this project does not justify the higher up-front capital costs associated with refrigeration based CO_2 liquefaction systems. Furthermore, solid desiccant dehydration systems such as activated alumina and molecular sieves are required prior to refrigeration based CO_2 liquefaction and the project did not otherwise require this additional level of dehydration with its associated higher capital and operating costs. There are cases where CO_2 liquefaction and pumping is the most cost-effective approach, however; the site-specific process conditions and goals for this project led the team to conclude that compression and triethylene glycol dehydration was the more economical and preferred alternative for this project.

The centrifugal compressor vendors that Trimeric contacted for budgetary quotes for this project did not provide budgetary quotes. It was difficult to get responses of any kind from these suppliers, but one did comment that this project is on the low end of flow rates where centrifugal compressors are applicable. Centrifugal compressors also have longer equipment delivery times and more limited turndown capabilities than reciprocating compressors. Due to these factors, centrifugal compressors were not considered further for this project.

PROCESS COOLING OPTIONS

Process cooling for this project could have been achieved by either air coolers (fin-fan coolers) or by shell and tube heat exchangers utilizing cooling tower water. A comparison of the two showed that shell and tube exchangers with cooling water option would 1) provide tighter process temperature control within the surface facilities equipment 2) be more conducive to the planned indoor installation of the process equipment, and 3) have a smaller overall footprint, which was critical for this installation within the existing ADM facility. From a cost and performance standpoint, the air-cooled and water cooled options are comparable, with a slight advantage for the water cooled shell and tube heat exchangers due to the lower interstage cooling and final compressor discharge temperatures that can be achieved on a warm summer day. In many cases, air cooling is the preferred option if cooling water is not available or where water resources are limited. Cooling water from an existing cooling water system was available for use on this project.

DEHYDRATION

The basis for setting the gas water content specification at 30 lbs/MMscf (633 ppmv) was that it is typical of what might be specified for a commercial CO_2 pipeline transporting CO_2 for enhanced oil recovery or sequestration

purposes. Dehydration prevents liquid water and/or hydrates from forming and helps to prevent internal corrosion of the 6,400-foot carbon steel pipeline. Both Triethylene Glycol, TEG, and solid desiccant (molecular sieve) dehydration systems were evaluated. A TEG dehydrator will normally dehydrate a CO_2 stream to a water content of 7 lbs/MMscf or lower. A molecular sieve system typically dehydrates to an even lower water content, less than 1 lb/MMscf. The option of TEG dehydration was selected as it provided an acceptable level of dehydration for the project requirements and provided a good safety margin below the project design basis requirements. The project requirements did not justify higher capital and higher operating costs associated with molecular sieve dehydration.

4.2 Equipment Selection

During the equipment selection phase of the project, injectivity testing of the formation had yet to be conducted and as noted earlier, the required wellhead inlet pressure was unknown. It was expected to fall within a range from 1,400 to 1,950 psig. Thus, the selected compression equipment had to have a great deal of operational flexibility in order to meet the project requirements for injection rate and injection rate accuracy over the range of possible required surface injection pressures.

The results of the equipment screening process indicated that the compression configuration that had the lowest purchased equipments costs and 3-year electrical costs was the combination of one blower, two screw compressors, one reciprocating compressor, and one pump. However, based on input from equipment suppliers during the firm equipment bid solicitation process, Trimeric and ADM concluded that the costs for the option with one blower, two reciprocating compressors, and one pump were comparable to the screw compressor option. The total estimated purchased equipment costs combined with 3-year electrical costs for the two options were within 5% of each other for these two options. Eliminating the screw compressors reduced the overall complexity of the system and the number of large equipment skids. Therefore, the option that included a blower, two reciprocating compressors and a pump was chosen.

The reciprocating compressors were designed to operate at a discharge pressure of 1,400 psig followed by a multistage centrifugal pump if additional pressure is needed to meet the desired injection rate. This option provided the needed flexibility for the possible range of surface injection pressures and avoided having to operate larger reciprocating compressors, designed for meeting discharge pressures up to 2,000 psig, in a constant and significant turndown situation if a lower injection pressure was required to meet the desired injection rate.

The selected process is depicted in the Process Block Blow Diagram (

Figure 2) and the Process Flow Diagram (Figure 3). Several photographs of the system are located in Appendix 1.

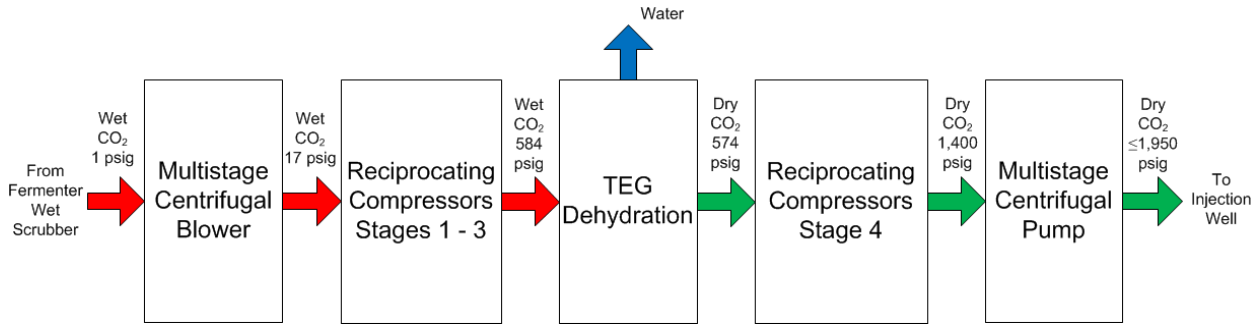


Figure 2 – Process Block Flow Diagram

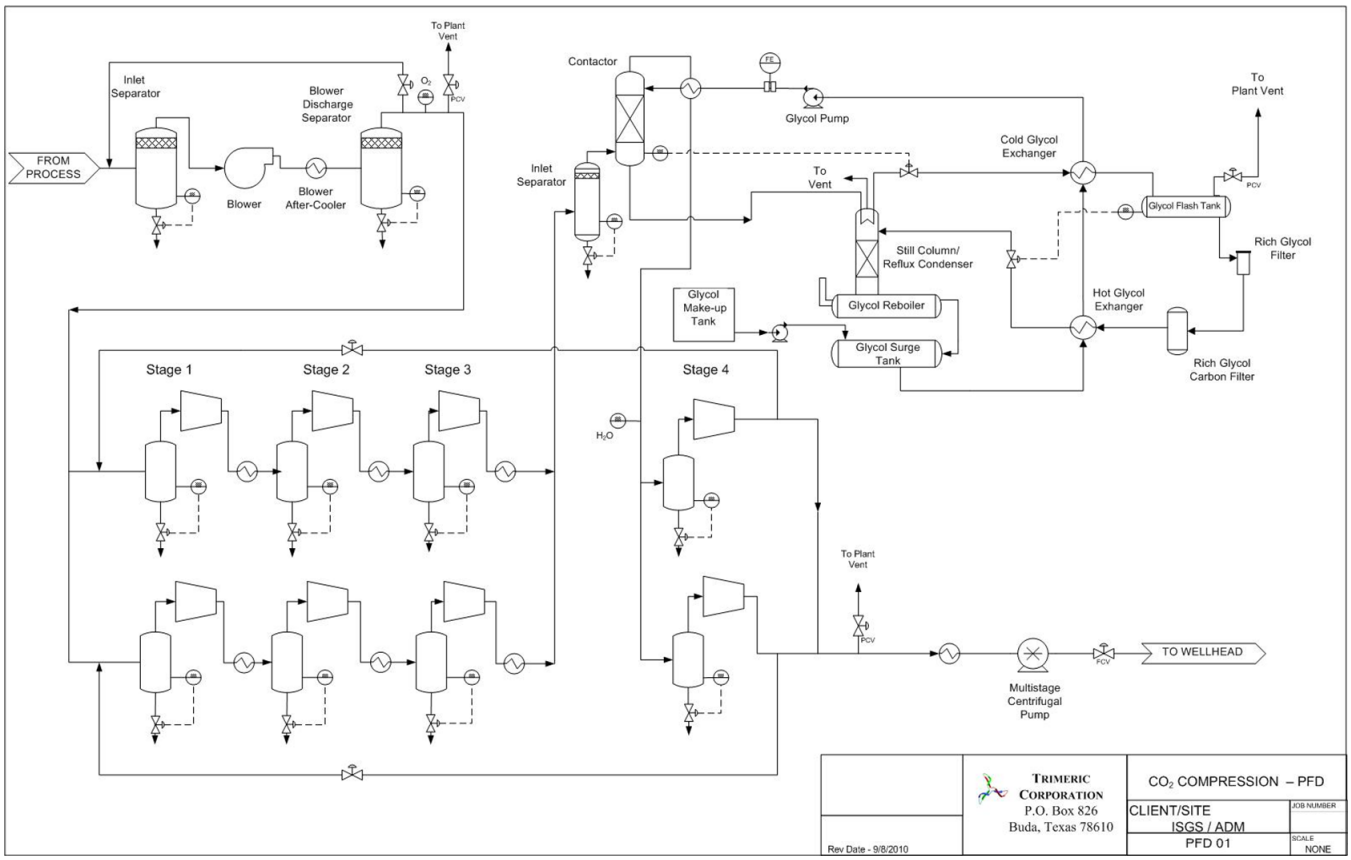


Figure 3 – Surface Facilities Process Flow Diagram

The selected surface facility equipment consists of a single 1,250 hp 4-stage centrifugal booster blower that raises the CO_2 stream pressure to approximately 17 psig. It is followed by two parallel 3,250 hp 4-stage reciprocating compressors that boost the pressure to approximately 1,400 psig, a TEG dehydration unit, and a 200 hp 26-stage centrifugal pump capable of boosting the pressure to up to 2,000 psig. TEG dehydration is performed between the third and fourth stages in the reciprocating compressors where water content in the CO_2 is at a minimum due to previous compression and cooling steps. Shell and tube heat exchangers using cooling water remove the heat

of compression following each compression step, except after the multistage centrifugal pump, which adds a temperature rise of only 10 ^{o}F to 15 ^{o}F and therefore does not justify addition of a cooler after the pump. Additional description of the Surface Facilities is provided in the following paragraphs.

The compression train receives the relatively low pressure CO_2 stream from the existing primary water scrubber overhead stream. The gas enters the inlet separator where any free water carry-over from the scrubber is allowed to drop out. The stream enters the blower where the CO_2 pressure is increased by approximately 17 psi. The gas stream is monitored for the presence of oxygen by an online oxygen analyzer; a high oxygen reading may indicate an air leak into the system. The gas stream is then cooled by a shell and tube gas cooler. The outlet gas temperature is controlled at a set point of 95 ^{o}F.

The CO_2 stream then exits the blower after cooler separator where any condensed liquid is allowed to drop out. The overhead stream from the blower after-cooler separator is split and then enters the suction of two parallel 4-stage reciprocating compressors.

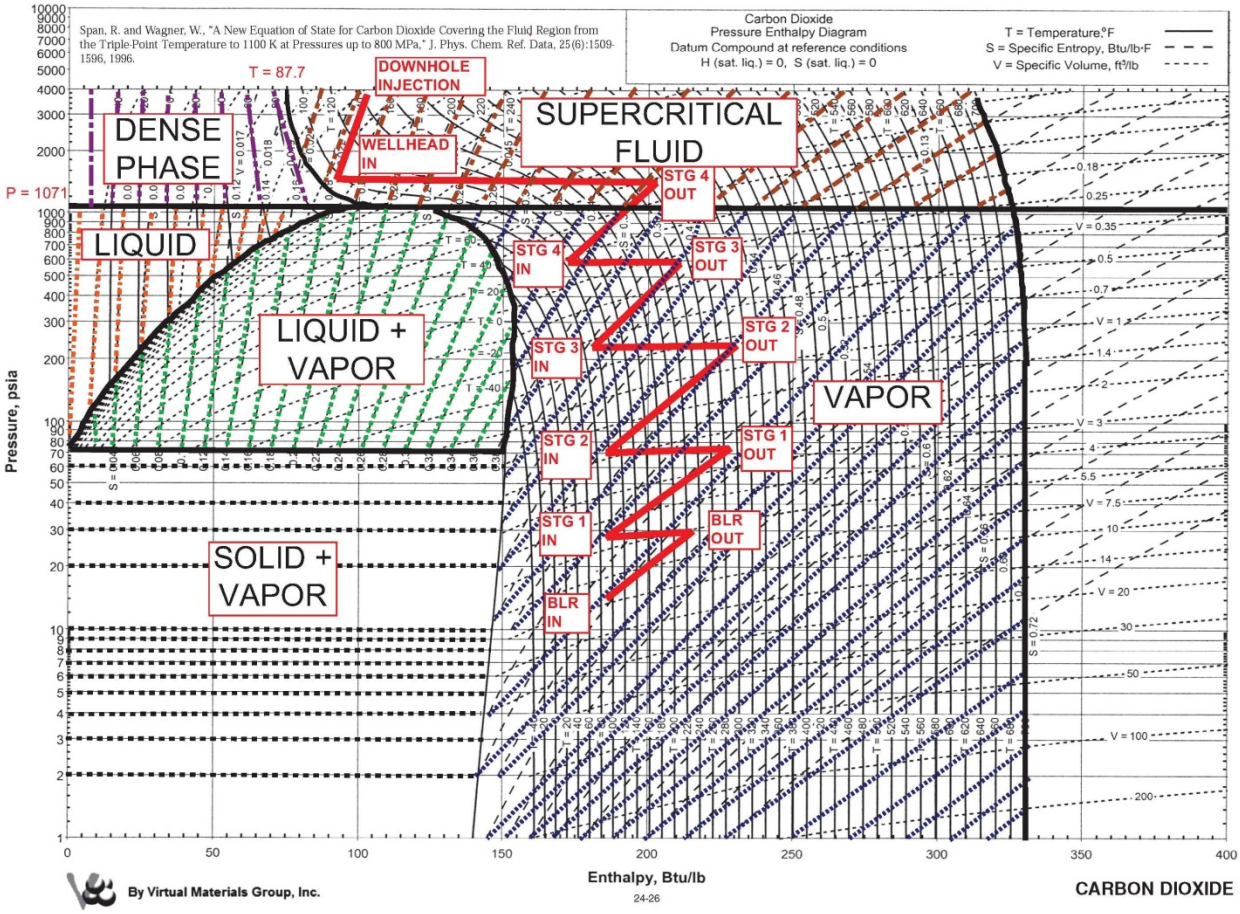
Compressed CO_2 from the 3rd stage of reciprocating compressors combines and enters the TEG dehydration unit inlet separator where condensed liquids disengage from the vapor stream. The CO_2 stream then enters the bottom of the contactor where it is contacted counter currently with the lean (low water content) TEG that enters at the top of the contactor. The dry CO_2 exiting the top of the contactor is cooled in a gas/glycol exchanger before splitting and returning to the 4th stage compressor suction scrubbers. The water content of the dehydrated gas stream is measured by an online moisture analyzer; a high moisture content reading may indicate an operational issue in the dehydration unit. Water is evaporated from the rich TEG in the TEG dehydration unit regeneration skid and the lean TEG is recycled to the contactor.

Figure 4 shows the path of the compression on a CO_2 Pressure-Enthalpy Phase Diagram. Note that the selected pressures and temperatures throughout the system remain outside the liquid, liquid/vapor, and solid/vapor envelopes, where serious operational issues would occur. CO_2 temperatures also remain well above 55 ^{o}F, where it is possible to form CO_2-H_2O solid hydrates under some conditions.

The compressor discharge pressure after the 4th stage is controlled at set point by a control valve that allows excess compressed CO_2 to flow to the process vent header if the injection rate to the wellhead is reduced.

Temperature control loops for each of the interstage shell and tube exchangers (three for each compressor) as well as the temperature control for the final after cooler will control the outlet CO_2 temperature at a set point of 95 ^{o}F. At this point, the CO_2 can bypass the multistage centrifugal pump if it is not needed or alternatively go through the pump if it is needed to meet the desired injection rate.

CO_2 flow to the wellhead is monitored by a flow indicating transmitter and is controlled by a flow control valve.



Pressure-Enthalpy Phase Diagram used with Permission of Gas Processors Suppliers Association

Figure 4 – Compression Path on CO_2 Phase Diagram

5.0 Results and Lessons Learned

The system capacity was designed for 1,100 tonnes / day, which provides a 10% safety factor above the desired 1,000 tonnes / day injection rate. The average rate leaving the compression facility during normal operations to date has been 1,086 tonnes / day. Figure 5 shows a plot of the blower capacity curve and the reciprocating compressor curves provided by the equipment suppliers and their expected intersection point at 1,103 tonnes / day leaving the compression facility. Actual operating data collected during the commissioning effort showed very good agreement with these operating curves and their intersection point

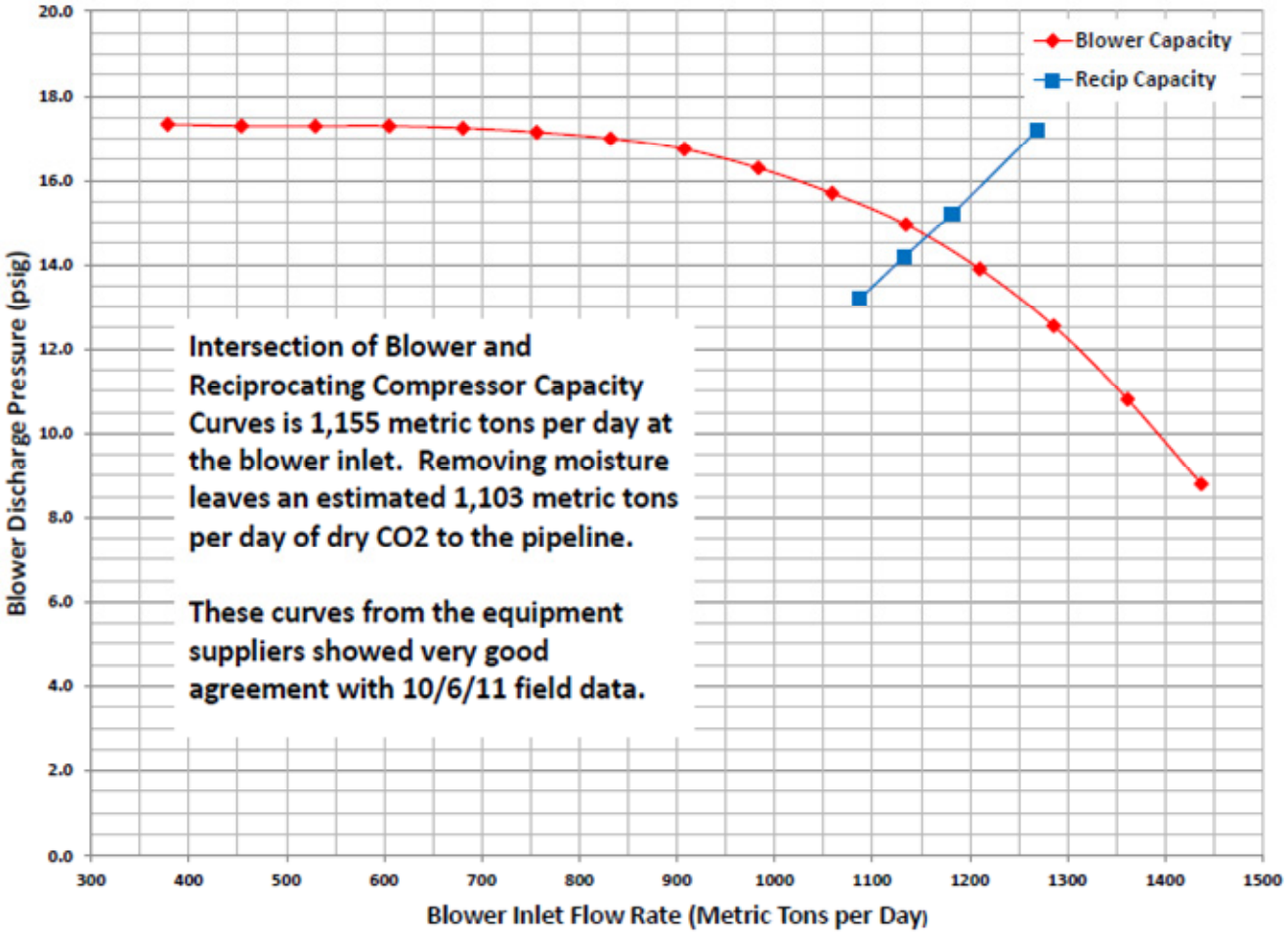


Figure 5 – Intersection of Blower and Reciprocating Compressor Curves

As shown in Figure 6 the average injection rate during normal operations to date has been 974 tonnes / day. ADM is working with Illinois EPA on an administrative issue to remove a current permit constraint that limits the injection rate based on the size of the orifice meter plate in use. After this is resolved, the injection rate will increase by about 100 tonnes / day.

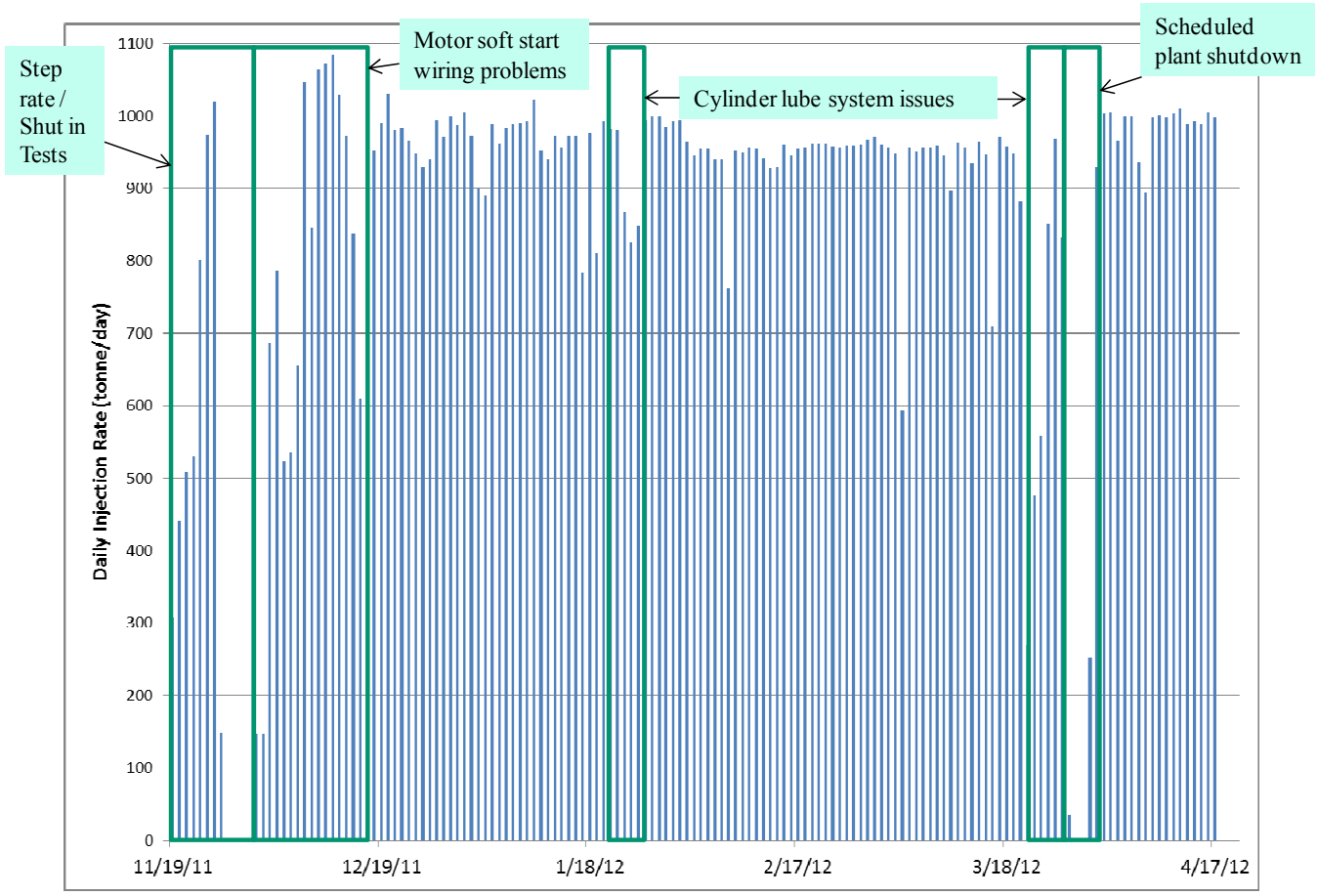


Figure 6 – Daily Injection Rates

In the first six months of operation there have been some issues that led to unscheduled downtime. There has also been one scheduled maintenance period for the compression equipment that coincided with a scheduled ethanol plant shutdown. Main issues leading to unscheduled downtime to date are as follows:

- Blower motor failure – windings arced / phase to phase ground due to poor insulation (8 days)
- Motor soft start wiring issues (9 days)
- Cylinder lubrication issues (rupture disk and pump issues) (8 days)
- Ethanol plant outages - cooling water, power, CO_2 supply (3 days)

The blower motor failure occurred in early Nov. 2011 before injection began in earnest. It was a warranty issue that was repaired by the supplier at no cost to ADM or the project. The impact of the other scheduled and unscheduled downtime events is shown on Figure 6. ADM is working with the equipment packager and the compressor manufacturer to improve the reliability of the cylinder lubrication systems. Unscheduled downtime is expected to decrease now that initial issues have been resolved.

LESSONS LEARNED

The project site is located within the existing ADM Decatur, IL Corn facility. Due to the existing facility infrastructure, it was not practical to install the 6,400 foot, 6 inch diameter pipeline underground. ADM initially operated without insulating the above ground pipeline to evaluate the need for insulation. Ambient conditions

affected CO_2 temperature and therefore density at wellhead inlet, which reduced the surface pressure required for a given injection rate in colder weather. As a result, compression equipment control set points had to be modified to operate with lower discharge pressure. Furthermore, variations in ambient conditions caused undesirable variations in subsurface injection pressures. Therefore, the pipeline was insulated in order to reduce impacts of ambient conditions on subsurface pressure and to ensure that wellhead inlet temperature will remain above 60 ^{o}F minimum limit in the UIC injection permit.

Trimeric modeling based on initial injection operations predicts that a surface pressure of 1,455 psig will be required to inject 1,000 tonnes / day if surface CO_2 temperature is 95 ^{o}F. Project geologists and reservoir engineers expect the required injection pressures to decline as injection continues (with respect to the injection pressures encountered during initial operations), but if required injection pressures do not decline and if the performance of the final cooler after the compressor degrades due to fouling or other reasons, the multistage centrifugal pump may be needed to achieve desired injection rates in warmer weather when cooling water supply temperatures to the heat exchangers are higher.

It is sometimes difficult to respond to startup, non-steady-state, and upset conditions using the vent valve downstream of the blower and the vent valve downstream of the reciprocating compressors. The vent valve after the reciprocating compressors was difficult to tune during commissioning efforts before injection into the well and therefore use of the pipeline was permitted. After the injection permit was granted and injection began, additional tuning of the vent valve downstream of the compressor combined with the dampening effect the 6,400 foot, 6 inch pipeline had on changing conditions, and widening the compressor discharge pressure shutdown set points, has resulted in satisfactory operation of the vent valve downstream of the compressor. The single blower is sized to feed two reciprocating compressors in parallel. Consequently, when one compressor shuts down, the vent valve downstream of the compressor has to respond quickly to vent half of the gas going through the blower in order to reduce pressure at the discharge of the blower and at the suction to the remaining on-line reciprocating compressor in order to avoid a shutdown of the remaining on-line reciprocating compressor due to high suction pressure or a shutdown of the blower due to high discharge pressure. On the other hand, if the vent valve downstream of the blower opens too quickly, the other reciprocating compressor will shut down on low suction pressure and / or the blower will shut down on low blower discharge pressure. Opening of the blower recycle valve in addition to opening of the vent valve downstream of the blower has reduced the number of compressor shutdowns that occur when the other compressor shuts down. The blower usually remains on-line when one or both compressors shuts down. Keeping the remaining units on line when one compressor shuts down may reduce the time required to get back to the full injection rate because the motor soft starts can impose a one-hour delay before a motor can be restarted. Shutdown limits for compressor suction pressure will also be widened to reduce the chances of one compressor shutting down as a result of the other compressor shutting down. It is possible that additional valve tuning may be required for the vent valve downstream of the blower and the vent valve downstream of the reciprocating compressors.

Operational changes were needed for lower than design cooling water supply temperatures. The system was designed based on a summer design cooling water supply temperature of 85 ^{o}F. During the initial months of operation, cooling water temperatures have been lower. The flow rate of cooling water through the heat exchangers after the blower and reciprocating compressors is adjusted to achieve a desired discharge temperature from these exchangers. The initial target temperature for CO_2 leaving these exchangers was 95 ^{o}F, which is a 10 ^{o}F approach for the summer design cooling water supply temperature of 85 ^{o}F. When the cooling water supply temperature is lower, the flow rate of cooling water to the exchangers is reduced which results in high cooling

water discharge temperature from the heat exchangers. Higher cooling water discharge temperatures increase the risk of fouling and corrosion in the cooling water system and in the heat exchangers. Getting the CO_2 too cold (below roughly 60 ^oF) could lead to issues with CO_2-H_2O solid hydrate formation or formation of liquid CO_2 at pressures in the latter stages of the reciprocating compressors. ADM lowered the CO_2 temperature set points to 80 ^oF for interstage cooling in order to get cooling water flow rate up and therefore to reduce the cooling water discharge temperatures. ADM then had to lower interstage low suction pressure shutdown set points as suction pressure was lower with cooler (denser) CO_2. ADM may decide to add a cooling water tempering loop to get cooling water supply temperature up in cold weather. The tempering loop would pump some of the cooling water return from the facility to the cooling water supply in order to raise the blended temperature of the cooling water going to the heat exchangers in this facility.

6.0 Comparison with 500 MWe Facility

The 1,000 tonnes per day CO_2 injection rate in the Illinois Basin – Decatur Project is about 10% of the CO_2 that is emitted from a 500 MW (gross) PC power plant. A facility 10 times larger such as a 500 MW PC power plant would mostly likely use centrifugal compressors instead of reciprocating compressors. While larger capacity, lower speed (more reliable and more expensive) reciprocating compressors than those used on the IBDP are available, the higher reliability and still higher per unit throughput capacities of centrifugal compressors compared with reciprocating compressors are very likely to favor selection of centrifugal compressors for a 500 MW PC power plant CO_2 sequestration application.

Triethylene glycol (TEG) dehydration technology is commonly used to dehydrate CO_2 that goes into commercial CO_2 transport pipelines in the U.S. Commercial systems with the capacity to dehydrate the flow rate of CO_2 that could be captured from a 500 MW PC power plant are available and in use at facilities that process CO_2 for enhanced oil recovery (EOR) in the U.S. These units may require multiple trains of glycol contactors for the 500 MW PC power plant scale. It is unlikely that the higher capital operating costs of mole sieve dehydration would be warranted for the power plant sequestration application. TEG dehydration is sufficient and the higher levels of dehydration are not required in this application. It may be possible in some instances to get by with less dehydration and therefore lower cost than TEG dehydration. The most common approach in this instance is to design the injection pipeline and injection well to accommodate the water that remains in the CO_2 after compression and cooling steps. Applications with shorter distance from the compression facility to the injection well and those that do not require meeting a commercial CO_2 pipeline specification for water content may most easily accommodate this approach. However, higher grade – higher cost materials for the pipeline and injection well may be required. Non-steady state conditions such as extended shutdowns must be considered with respect to the possibility for free water (liquid water) formation which can lead to accelerated corrosion rates in CO_2 service and for the possibility of CO_2-H_2O solid hydrate formation which could lead to plugging in the injection well perforations.

Innovative technologies are in development to reduce compression and dehydration costs at the scale required for carbon capture for a 500 MW PC power plant. Examples include the Ramgen[1,2] compressor and the DexPro™ dehydration process. [3,4]

7.0 Summary and Conclusions

The compression and dehydration system for the Illinois Basin-Decatur Project performs in accordance with process design. The flow rate of CO_2 from the system that is available for injection meets the design criteria of 1,100 tonnes per day. Dehydration levels are typically in the 5 lb/MMscf range, as expected for performance of this system. System availability is expected to increase and unscheduled downtime is expected to decrease now that the first few months of operation are complete and the issues impacting system reliability have been identified. Most of these issues were resolved during the first few weeks of operation. ADM is working with equipment suppliers to improve reliability of the cylinder lubrication systems on the reciprocating compressors.

Further tuning of pressure vent valves downstream of the blower and downstream of the reciprocating compressors may be needed to improve the response of the system to upset conditions such as the response of the vent valve downstream of the blower when one of the two reciprocating compressors shuts down. Further planned adjustments of suction pressure set points for compressor shutdown will also improve the system response to upset conditions.

ADM may decide to install a tempering loop to raise the cooling water supply temperature in cold weather. However, lowering the target CO_2 discharge temperature from the heat exchangers from 95 ^{o}F to 80 ^{o}F may be sufficient to achieve desired cooling water return temperatures without adding a cooling water tempering loop.

The pipeline has been insulated, which will ensure that the CO_2 delivered to the wellhead inlet will remain above the minimum injection temperature of 60 ^{o}F allowed in the UIC permit and will greatly reduce the impact of ambient conditions including temperature, wind, and precipitation on the temperature and density of the CO_2 delivered to the wellhead. This in turn will reduce variations in down-hole pressure and temperature associated with surface facility operations. Use of the multistage centrifugal pump may be needed to maintain desired injection rates in warmer weather. Since there is not a cooler after the pump and since the pump will add 10 ^{o}F to 15 ^{o}F to the CO_2, insulating the pipeline increases the chance that the pump will need to be used in warm weather to meet the desired injection rate as the warmer CO_2 delivered to the wellhead will be less dense.

Quarterly performance monitoring is underway to track, maintain, and improve system performance over the remainder of the injection period. Trimeric has made and plans to continue making quarterly visits to the site during the injection period of this project to review operations, to identify process issues, to identify opportunities for process optimization, and to support ADM and the Illinois State Geological Survey during the injection phase of the Illinois Basin-Decatur Project.

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9.0 Appendix

Appendix 1 – Surface Facilities Equipment Photos

APPENDIX 1



Photograph 1 – Blower and Compressor Buildings



Photograph 2 – CO_2 Source – Fermenter Wet Scrubber Discharge



Photograph 3 – Blower System



Photograph 4 – Reciprocating Compressor (Stages 1 and 4)



Photograph 5 – Reciprocating Compressor (Stages 2 and 3)



Photograph 6 – Interstage Coolers



Photograph 7 – TEG Dehydration System Contactor



Photograph 8 – TEG Dehydration Regeneration Skid



Photograph 9 – Final Compressor Aftercooler



Photograph 10 – Multistage Centrifugal Pump



Photograph 11 – CO_2 Injection Well

References Cited

[1] Baldwin, P., Williams, J., "Capturing CO_2: Gas Compression vs. Liquefaction", *Power Magazine*, published online 01 June 2009.
[2] "A Shocking Way to Compress CO_2", *Chemical Engineering*, June 2009.
[3] McKay, W., Maddocks, J., Hansen, V., "CO_2 Dehydration: Why? How Much? How? ", Tenth Annual Conference on Carbon Capture and Sequestration, May 2 – 5, 2011.
[4] "CO_2 Dehydration for Pipeline Safety", *Carbon Capture Journal*, Transport and Storage, September/October 2011, Pages 22-24.